

Distributed AI Agents for Cognitive Underwater Robot Autonomy



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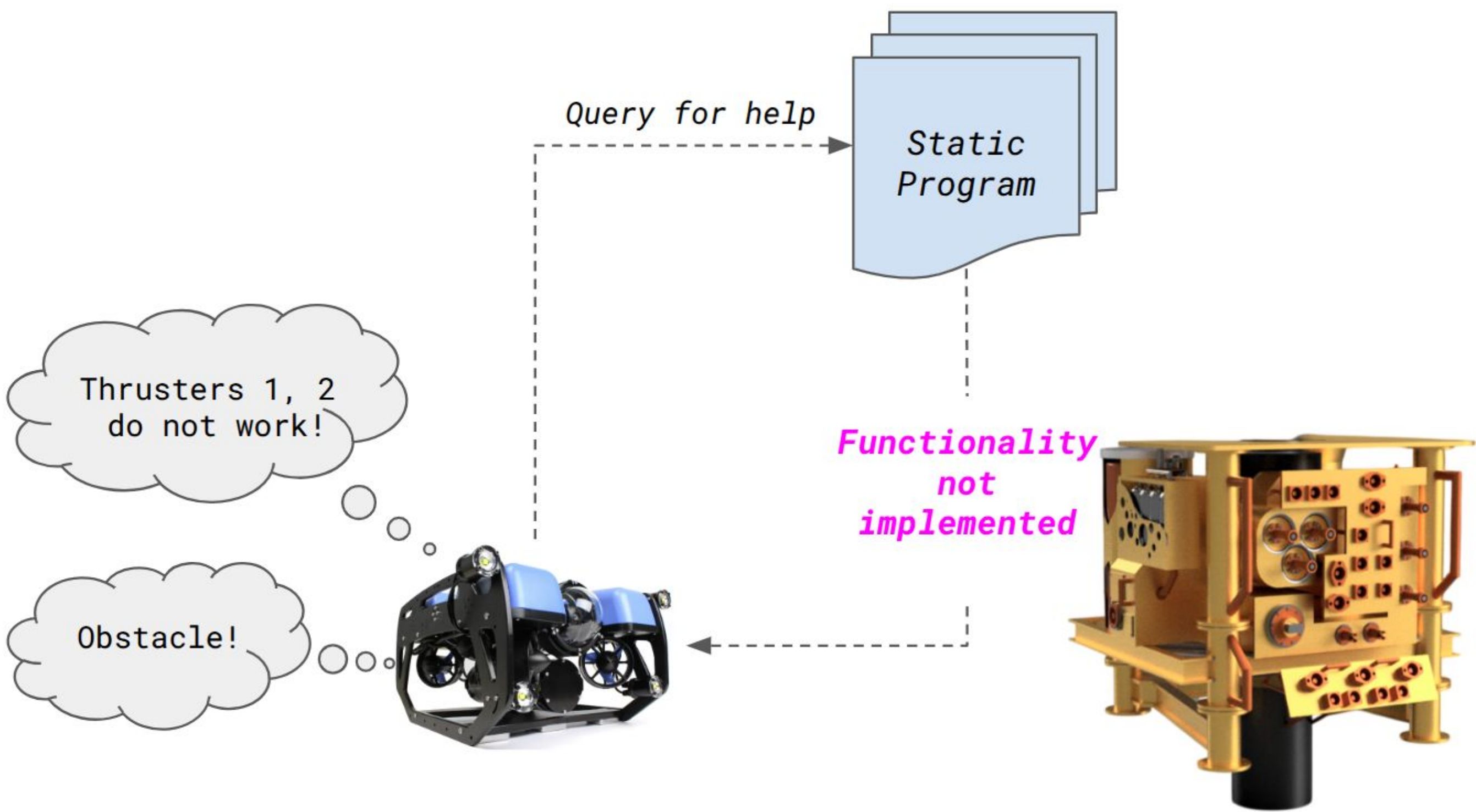
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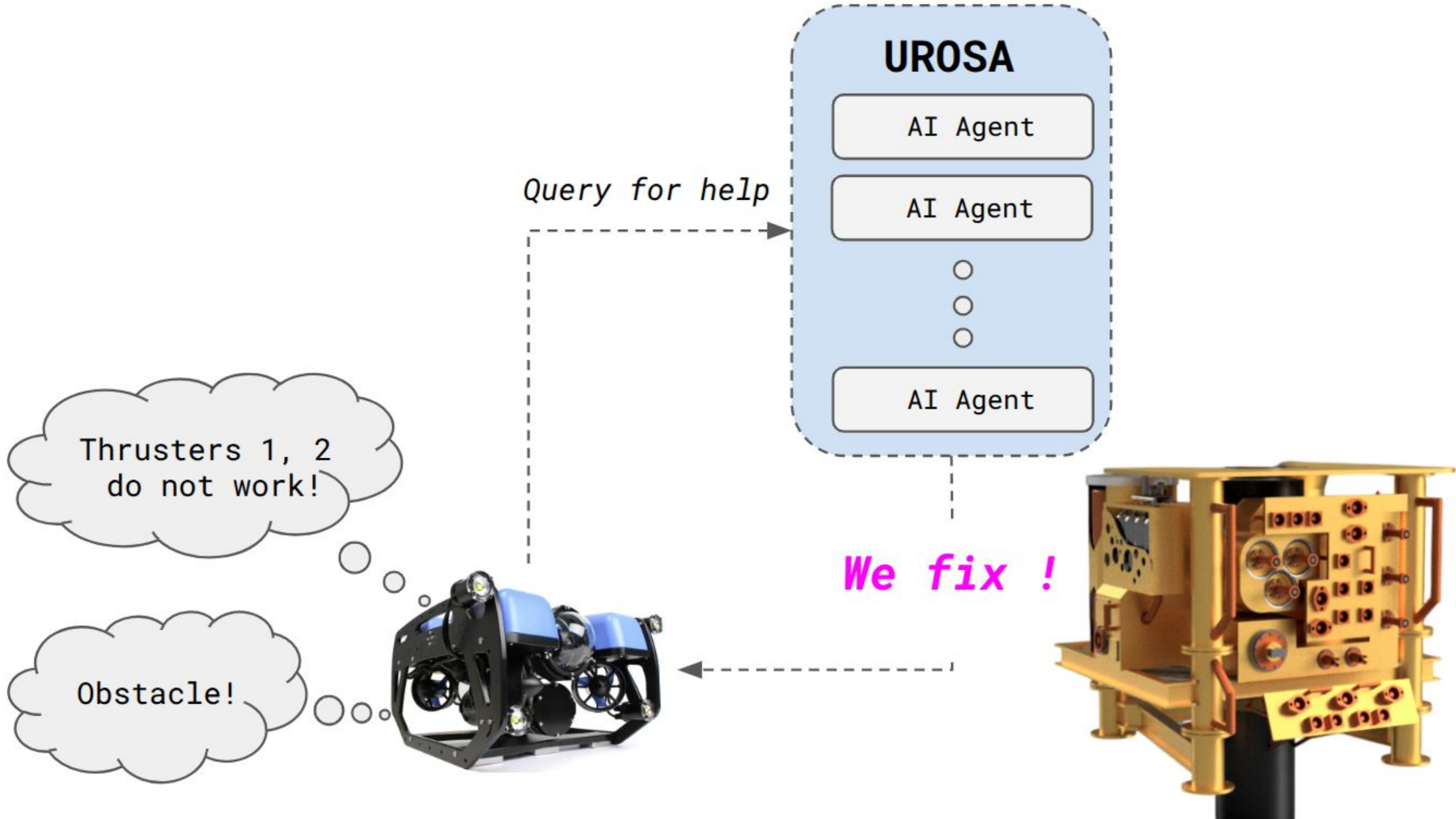
MOTIVATION

Traditional underwater robots struggle with uncertainty and failure. UROSA enables cognitive autonomy through distributed AI agents capable of reasoning, learning, self-diagnosis, self-repair, and autonomous software generation.

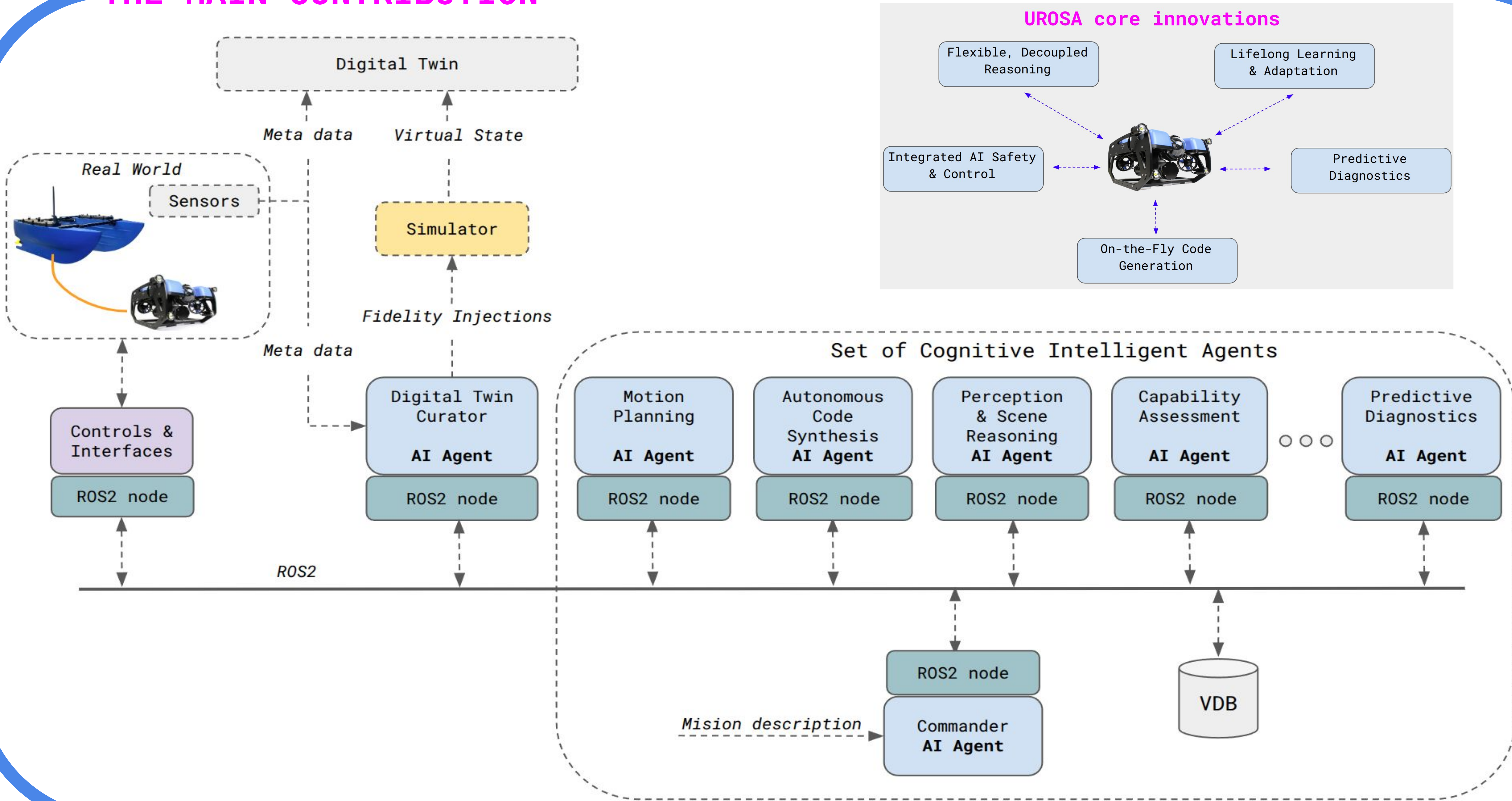
Traditional Robotics



OURS

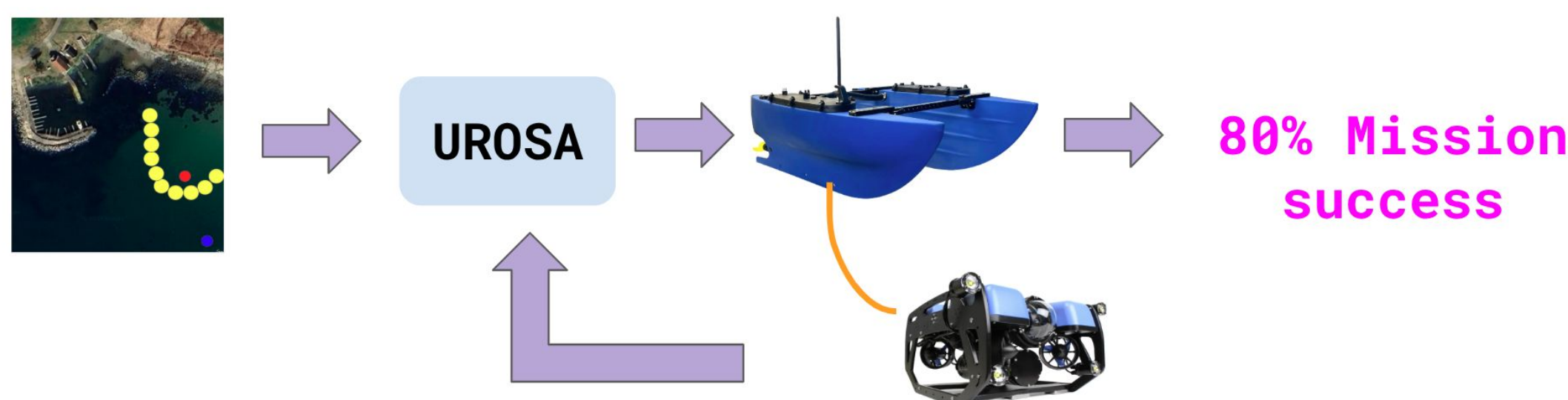


THE MAIN CONTRIBUTION

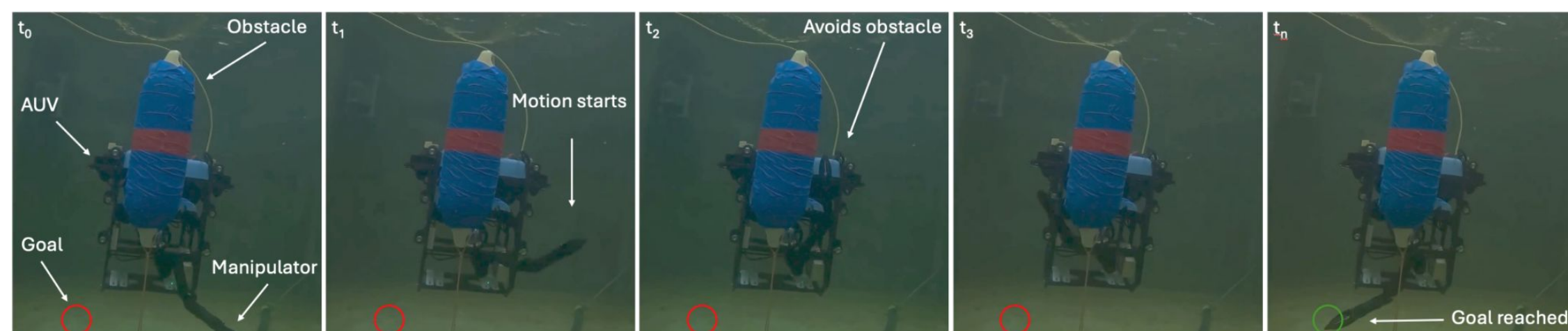


KEY RESULTS

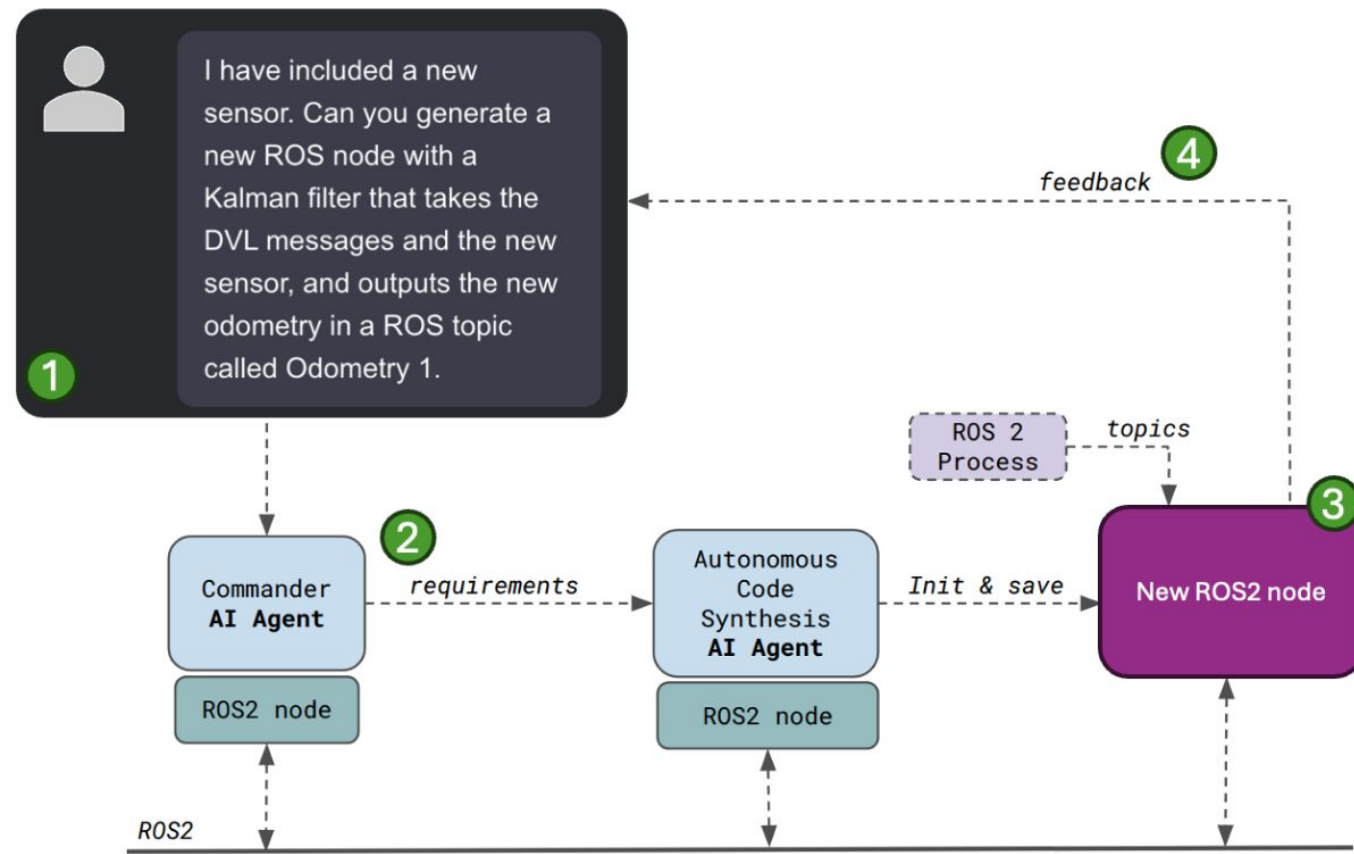
80% Mission success in complex perception-driven multi-robot navigation.



100% Planning success once task intent is correctly understood.



70% Reduction in navigation drift through autonomously generated repair software.



100% Diagnostic accuracy for tested thruster fault scenarios.

